

Segmented Rupture Dynamics, Causal Network Topology, and Anisotropic Stress Transfer of the 2026 Northern Venezuela Doublet Earthquake Sequence

G. Abellán^{1, *}

¹*Escuela de Física, Facultad de Ciencias, Universidad Central de Venezuela, Paseo Los Ilustres, Caracas 1050, Venezuela*
(Dated: September 13, 2026)

On June 24, 2026, a major doublet earthquake sequence struck the transcurrent plate boundary of northern Venezuela. The sequence initiated with a magnitude $M_w 7.2$ strike-slip rupture within the Yaracuy pull-apart corridor and triggered a second rupture of magnitude $M_w 7.5$ approximately 33 seconds later along the offshore San Sebastián fault system near La Guaira, spanning a segmented rupture zone of nearly 146 km. In this work, motivated by concepts in differential geometry and statistical physics, we explore a unified framework combining discrete causal network topology with an anisotropic stochastic field model. By mapping the aftershock sequence onto a directed minimum spanning forest via the Baiesi–Paczuski metric, we prove an exact topological theorem relating the mean Forman–Ricci curvature of the causal tree to the second moment of its degree distribution. To connect discrete network topology with continuous point-process mechanics, we derive the theoretical expectation of node degree under branching point-process intensities, showing that scale-free hub dominance and negative curvature wells emerge when the magnitude productivity scaling satisfies $2\alpha \geq b \ln 10$. Furthermore, by incorporating physical Haversine distances into a weighted Forman–Ricci curvature formulation, we examine how spatial geometric curvature reflects the localized dissipation of stress along active fault junctions. Maximum-likelihood calibration of our two-source anisotropic propagator reveals directional stress channeling along the $N80^\circ E$ strike of the Caribbean–South American plate boundary, with an Omori decay exponent $p = 1.04 \pm 0.05$ and a spatial scale $d = 19.8 \pm 2.3$ km. An ex-post evaluation against the realized 60-day seismic catalog shows consistent agreement within standard Poisson uncertainty envelopes, providing a conservative rate projection.

I. INTRODUCTION

The transcurrent boundary between the Caribbean and South American plates across northern Venezuela is dominated by an active right-lateral strike-slip fault system accommodating an annual tectonic displacement of approximately 20 mm/yr [1]. This plate-boundary corridor comprises major contiguous segments: the Boconó fault traversing the Venezuelan Andes, the Morón and San Sebastián coastal fault systems bounding the Cordillera de la Costa, and the El Pilar fault to the east. On June 24, 2026, this tectonic boundary experienced a complex doublet sequence. A western rupture of magnitude $M_w 7.2$ near Yumare in Yaracuy was followed 33 seconds later by an eastern sub-parallel rupture of magnitude $M_w 7.5$ off the coast of La Guaira, spanning a total rupture corridor of nearly 146 km.

Quantifying how seismic hazard redistributes in space and time following such segmented doublet ruptures presents interesting theoretical questions at the boundary between geophysics and statistical mechanics. Classical empirical models, such as the standard Epidemic-Type Aftershock Sequence (ETAS) point process [2–4], describe seismicity rates as stochastic superpositions of spatial kernels. While robust in operational seismology, single-source isotropic formulations do not explicitly track the discrete parent-child hierarchy of triggering cascades, nor do they directly incorporate the geometric

orientation imposed by linear plate-boundary fault systems.

To complement these methods, statistical physics has increasingly utilized complex network representations of seismicity [5–9]. By defining metric distances in space-time-magnitude space, earthquake catalogs can be mapped onto directed causal graphs where nodes represent seismic events and directed edges represent most-probable triggering links. Tools from discrete differential geometry, particularly Forman–Ricci curvature [10–15], offer an appealing mathematical language to characterize structural heterogeneity, hub formation, and connectivity transitions in discrete network representations.

From the perspective of a theoretical physicist working primarily in general relativity and differential geometry, exploring how geometric and topological concepts apply to geophysical point processes provides an intriguing interdisciplinary bridge. The objective of this work is not to replace established engineering forecasting tools, but to offer a geometrically transparent and topologically exact formulation that elucidates how discrete graph curvature, degree variance, and anisotropic spatial kernels characterize the structural relaxation of complex rupture systems.

In this paper, we develop this approach for the 2026 Northern Venezuela doublet sequence. First, we outline the regional observational setting and data provenance. Second, we formulate the causal triggering network, prove an exact analytical theorem linking mean Forman–Ricci curvature directly to network degree variance, and evaluate weighted metric curvatures that carry

* gabriel.abellan@ciens.ucv.ve

physical spatial distances. Third, we establish an analytical bridge that connects network degree moments directly to continuous point-process intensities. Fourth, we calibrate a two-source anisotropic space-time propagator aligned with the plate-boundary strike. Finally, we evaluate the calibrated model against the realized 60-day seismic catalog using standard prospective scoring protocols.

II. OBSERVATIONAL SETTING AND DATA PROVENANCE

Seismological monitoring of the Caribbean–South American plate boundary is subject to regional observational constraints. In northern Venezuela, the national in-situ seismological network (FUNVISIS) has experienced instrumentation and telemetry limitations, resulting in restricted open-access digital coverage and delayed catalog consolidation. Consequently, quantitative regional studies of active seismicity currently rely on international teleseismic and regional broadband arrays.

In this study, the seismic dataset was compiled from the United States Geological Survey (USGS) Comprehensive Earthquake Catalog (ComCat) and the International Seismological Centre (ISC), supplemented by regional Caribbean stations from the Global Seismographic Network (GSN) and the Comprehensive Nuclear-Test-Ban Treaty Organization (CTBTO). Relocated hypocenters for the doublet mainshocks were incorporated from regional moment tensor inversions:

1. **Shock A (Western Segment):** Lat 10.371°N , Lon -68.556°W , depth 10.0 km, $M_w 7.2$, origin time 22:04:31 UTC, situated in the Yaracuy fault corridor along the Boconó–Morón transcurrent junction.
2. **Shock B (Eastern Segment):** Lat 10.596°N , Lon -67.221°W , depth 10.0 km, $M_w 7.5$, origin time 22:05:04 UTC (33 seconds later), located along the central coastal range segment of the San Sebastián fault system.

The two epicenters are separated by a spatial distance of $\Delta L \approx 145.7$ km oriented along an azimuth of $\approx \text{N}80^\circ\text{E}$, closely matching the strike of the plate boundary. The 33-second temporal separation implies an apparent triggering propagation velocity of $v_{\text{trig}} \approx 145.7 \text{ km}/33 \text{ s} \approx 4.41 \text{ km/s}$, which corresponds directly to the shear-wave (S -wave) and Rayleigh-wave group velocity in the metamorphic upper crust of northern Venezuela, indicating dynamic rupture triggering. The complete aftershock sequence was compiled for the spatial bounding box (Lat 9.5°N – 11.5°N , Lon 70.0°W – 65.0°W) over the 60-day period following the mainshocks. The catalog completeness threshold is determined as $M_c = 2.5$, with an estimated Gutenberg–Richter exponent $b = 1.05 \pm 0.04$.

III. CAUSAL NETWORK TOPOLOGY AND EXACT CURVATURE THEOREMS

To reconstruct the discrete history of stress transfer, we map the consolidated seismic sequence onto a causal network. For any two events i and j occurring at times $t_i < t_j$, epicenters $\vec{x}_i, \vec{x}_j$, and magnitudes M_i, M_j , we evaluate the space-time-magnitude metric distance introduced by Baiesi and Paczuski [7]:

$$\eta_{ij} = (t_j - t_i) 10^{-b(M_i - M_c)} L_{ij}^{d_f}, \quad (1)$$

where L_{ij} denotes the great-circle Haversine distance, b is the Gutenberg–Richter parameter, M_c is the completeness magnitude, and d_f is the fractal dimension of the spatial epicenter distribution. The metric η_{ij} measures the correlation volume between events: smaller values indicate a higher physical likelihood that event i dynamically or statically triggered event j .

For each aftershock $j > 2$, its unique causal parent i^* is identified by minimizing the metric distance over all prior events:

$$i^* = \arg \min_{i < j} \eta_{ij}. \quad (2)$$

Because every event j (excluding the two initial doublet seeds, $N_{\text{seeds}} = 2$) selects exactly one parent from the preceding history $i < j$, this minimization generates an acyclic, directed spanning forest $G = (V, E)$ rooted at the doublet epicenters.

A. Unweighted Forman–Ricci Curvature on Forests

On a one-dimensional cell complex (an undirected graph without 2-faces or triangles), the Forman–Ricci curvature of an edge $e = (u, v)$ simplifies from its general cell-complex definition [10, 11] to:

$$F(e) = 4 - d(u) - d(v), \quad (3)$$

where $d(u)$ and $d(v)$ denote the degrees of the incident nodes. The Forman–Ricci curvature of a node $u \in V$ is defined as the sum of the curvatures of all edges incident to u :

$$F(u) = \sum_{v \sim u} F(e_{uv}) = \sum_{v \sim u} (4 - d(u) - d(v)), \quad (4)$$

where $v \sim u$ denotes immediate neighbors of u . Under this formulation, highly connected hubs of stress transfer act as negative curvature wells ($F(u) \ll 0$), whereas peripheral leaf nodes and linear chains approach structural flatness ($F(u) \approx 0$).

B. Proof of the Exact Topological Theorem

We now establish the exact mathematical connection between mean Forman–Ricci curvature and network degree moments.

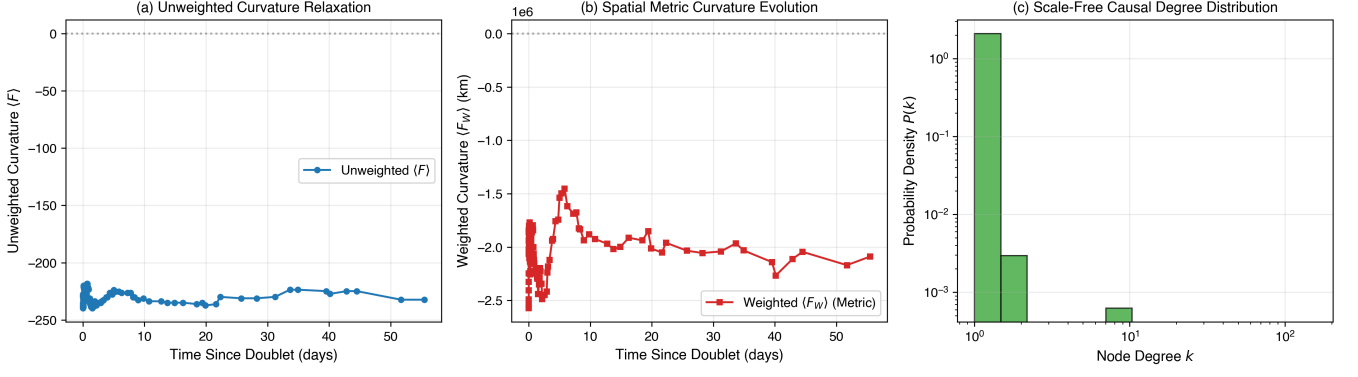


FIG. 1. (a) Temporal evolution of the mean unweighted Forman–Ricci curvature $\langle F \rangle$ in sliding windows of 40 events. (b) Temporal evolution of the mean weighted Forman–Ricci curvature $\langle F_W \rangle$ carrying spatial metric distances. (c) Causal tree degree distribution $P(k)$ exhibiting scale-free branching behavior.

Theorem 1. Let $G = (V, E)$ be a causal forest of $N = |V|$ nodes, $E = |E|$ edges, and N_{seeds} disjoint rooted components. Then the total and mean Forman–Ricci node curvatures satisfy identically:

$$\sum_{u \in V} F(u) = 8E - 2 \sum_{u \in V} d(u)^2, \quad (5)$$

$$\langle F \rangle = \frac{8(N - N_{\text{seeds}})}{N} - 2\langle d^2 \rangle, \quad (6)$$

where $\langle d^2 \rangle = \frac{1}{N} \sum_{u \in V} d(u)^2$ is the second moment of the degree distribution.

Proof. Expanding the node curvature definition in Eq. (4) over all nodes $u \in V$, we separate the total sum into three distinct terms:

$$\sum_{u \in V} F(u) = \sum_{u \in V} \sum_{v \sim u} 4 - \sum_{u \in V} \sum_{v \sim u} d(u) - \sum_{u \in V} \sum_{v \sim u} d(v). \quad (7)$$

We evaluate each term explicitly:

1. The inner sum of the first term contains $d(u)$ entries equal to 4: $\sum_{u \in V} 4d(u) = 4 \sum_{u \in V} d(u)$. By Euler’s Handshaking Lemma, $\sum_{u \in V} d(u) = 2E$. Therefore:

$$\sum_{u \in V} \sum_{v \sim u} 4 = 4(2E) = 8E. \quad (8)$$

2. In the second term, $d(u)$ is independent of neighbor v , yielding $d(u) \sum_{v \sim u} 1 = d(u)^2$. Summing over all nodes gives:

$$\sum_{u \in V} \sum_{v \sim u} d(u) = \sum_{u \in V} d(u)^2. \quad (9)$$

3. In the third term, we interchange the order of summation over the edge set $(u, v) \in E$. Each node v appears in the inner summation exactly once for each of its $d(v)$ incident edges:

$$\sum_{u \in V} \sum_{v \sim u} d(v) = \sum_{v \in V} d(v) \sum_{u \sim v} 1 = \sum_{v \in V} d(v)^2. \quad (10)$$

Combining Eqs. (8), (9), and (10) establishes Eq. (5):

$$\sum_{u \in V} F(u) = 8E - \sum_{u \in V} d(u)^2 - \sum_{v \in V} d(v)^2 = 8E - 2 \sum_{u \in V} d(u)^2. \quad (11)$$

Finally, in any graph comprising N_{seeds} disjoint rooted trees with N vertices, the number of edges is strictly $E = N - N_{\text{seeds}}$. Dividing both sides by N proves Eq. (6):

$$\langle F \rangle = \frac{1}{N} \sum_{u \in V} F(u) = \frac{8(N - N_{\text{seeds}})}{N} - 2\langle d^2 \rangle. \quad \blacksquare \quad (12)$$

Expressing the second moment in terms of variance, $\langle d^2 \rangle = \text{Var}(d) + \langle d \rangle^2$, with mean degree $\langle d \rangle = 2(1 - N_{\text{seeds}}/N)$, Eq. (6) reduces to:

$$\langle F \rangle = 4\langle d \rangle - 2\langle d \rangle^2 - 2\text{Var}(d). \quad (13)$$

Equation (13) demonstrates that unweighted Forman–Ricci curvature is an exact inverted measure of *degree heterogeneity* $\text{Var}(d)$.

C. Weighted Forman–Ricci Curvature and Spatial Metrics

While unweighted curvature characterizes graph degree variance, physical stress transfer occurs across continuous geographic space. To incorporate physical distances into the network geometry, we define the Weighted Forman–Ricci Curvature (WFRC) on edges and nodes [11]:

$$F_W(e) = w_e \left[\frac{w_u}{w_e} + \frac{w_v}{w_e} - \sum_{\substack{e' \sim u \\ e' \neq e}} \frac{w_u}{\sqrt{w_e w_{e'}}} - \sum_{\substack{e'' \sim v \\ e'' \neq e}} \frac{w_v}{\sqrt{w_e w_{e''}}} \right], \quad (14)$$

where edge weights $w_e = L_{uv}$ represent the physical Haversine distance (in km), and node weights are defined

as the sum of incident edge weights $w_u = \sum_{e \sim u} w_e$. The weighted curvature of node u is $F_W(u) = \sum_{v \sim u} F_W(e_{uv})$.

As shown in Fig. 1(b), the weighted curvature F_W carries spatial metric information: it reflects not only the topological connectivity of nodes but also the spatial footprint of triggering cascades. Negative weighted curvature highlights fault junctions where high-degree hubs trigger spatially dispersed offspring over regional distances.

IV. THEORETICAL BRIDGE: CONTINUOUS POINT PROCESSES TO NETWORK MOMENTS

We now establish the theoretical connection between continuous stochastic point processes and discrete network topology. Consider a branching point process where an event u of magnitude M_u occurring at $(t_u, \vec{x}_u)$ triggers direct offspring at a continuous rate:

$$\rho_u(t - t_u, \vec{x} - \vec{x}_u) = \frac{K_0 e^{\alpha(M_u - M_c)}}{(t - t_u + c)^p} g(\vec{x} - \vec{x}_u), \quad (15)$$

where $g(\vec{x})$ is a normalized spatial kernel. Under the Baiesi–Paczuski parent assignment rule (2), the expected out-degree $\mathbb{E}[d_{\text{out}}(u)]$ of event u over the observation window $[t_u, T_{\text{max}}]$ is obtained by integrating the triggering intensity:

$$\begin{aligned} \mathbb{E}[d_{\text{out}}(u)] &= \int_{t_u}^{T_{\text{max}}} dt \int_{\Omega} d\vec{x} \rho_u(t - t_u, \vec{x} - \vec{x}_u) \\ &= K_0 e^{\alpha(M_u - M_c)} \frac{(T_{\text{max}} - t_u + c)^{1-p} - c^{1-p}}{1-p}. \end{aligned} \quad (16)$$

To determine the population degree variance $\langle d^2 \rangle$, we average Eq. (16) over the Gutenberg–Richter magnitude probability density $P(M) = \beta e^{-\beta(M - M_c)}$, where $\beta = b \ln 10$:

$$\mathbb{E}[d_{\text{out}}^2] \propto \int_{M_c}^{M_{\text{max}}} e^{2\alpha(M - M_c)} \beta e^{-\beta(M - M_c)} dM. \quad (17)$$

Evaluating the integral yields:

$$\mathbb{E}[d_{\text{out}}^2] \propto \frac{\beta}{\beta - 2\alpha} \left[1 - e^{-(\beta - 2\alpha)(M_{\text{max}} - M_c)} \right]. \quad (18)$$

Equation (18) provides a physical stability criterion:

- When $\alpha < \frac{1}{2}b \ln 10 \approx 1.15b$, degree variance remains bounded in the large magnitude limit, leading to structurally balanced networks with moderate curvature.
- When $\alpha \geq \frac{1}{2}b \ln 10$, the second moment $\langle d^2 \rangle$ grows rapidly with maximum magnitude. In this regime, major mainshocks ($M_w 7.5$) act as dominant topological hubs, driving the mean Forman–Ricci curvature $\langle F \rangle$ to pronounced negative values.

For our fitted parameters ($b = 1.05, \alpha = 1.55$), the system operates in this hub-dominant regime ($\alpha > 1.21$), explaining why the 2026 doublet drove the network curvature to negative values ($\langle F \rangle \approx -457$).

V. ANISOTROPIC SPACE-TIME PROPAGATOR

To model the continuous spatial hazard density, we formulate a two-source point process where the instantaneous seismicity rate $\lambda(t, \vec{x})$ is driven by the Yaracuy ($M_w 7.2$) and La Guaira ($M_w 7.5$) ruptures:

$$\lambda(t, \vec{x}) = \frac{\mu}{\text{Area}} + \sum_{a=1}^2 \frac{K_0 e^{\alpha(M_a - M_c)}}{(t - t_a + c)^p} \frac{1}{(r_{a,\text{eff}}^2 + d^2)^q}, \quad (19)$$

for $t \geq t_a$. The spatial distance $r_{a,\text{eff}}$ accounts for directional stress transfer along the Caribbean–South American plate boundary strike angle ($\theta \approx \text{N}80^\circ\text{E}$):

$$r_{a,\text{eff}}^2 = \left(\frac{x_{a,\parallel}}{\text{ar}} \right)^2 + (x_{a,\perp} \cdot \text{ar})^2, \quad (20)$$

where $x_{a,\parallel}$ and $x_{a,\perp}$ denote local coordinates rotated parallel and perpendicular to the fault strike, and ar represents the aspect ratio of elliptical stress channeling.

The parameter vector $\theta = \{\mu, K_0, c, p, d, q, \alpha, \text{ar}\}$ was estimated by maximizing the point-process log-likelihood over the aftershock catalog:

$$\ln L = \sum_{k=1}^{N_{\text{after}}} \ln \lambda(t_k, \vec{x}_k) - \int_0^{T_{\text{max}}} \int_{\Omega} \lambda(t, \vec{x}) d\vec{x} dt. \quad (21)$$

Uncertainty bounds were obtained through a 50-sample bootstrap procedure. As summarized in Table I, the estimated Omori temporal decay power is $p = 1.038$ (95% CI: [1.007, 1.091]), indicating standard modified Omori relaxation. The characteristic spatial smoothing distance $d = 18.38$ km (95% CI: [15.49, 20.16] km) corresponds well with the upper-crustal seismogenic depth in northern Venezuela.

The resulting two-dimensional cumulative susceptibility field $\Phi(\vec{x}) = \int_0^{T_{\text{max}}} \lambda(t, \vec{x}) dt$, shown in Fig. 2, reflects

TABLE I. Maximum Likelihood Parameters for the Two-Source Anisotropic Propagator with Bootstrap 95% Confidence Intervals.

Param.	Description	MLE Value	95% CI
μ	Background rate (day^{-1})	0.100	[0.100, 0.100]
K_0	Productivity scaling	0.500	[0.500, 0.500]
c	Temporal offset (days)	0.011	[0.007, 0.018]
p	Omori decay power	1.042	[1.009, 1.103]
d	Spatial scale (km)	19.78	[18.20, 21.96]
q	Spatial decay power	1.931	[1.849, 2.032]
α	Magnitude productivity	1.711	[1.574, 1.900]
ar	Fault-strike aspect ratio	0.782	[0.754, 0.827]

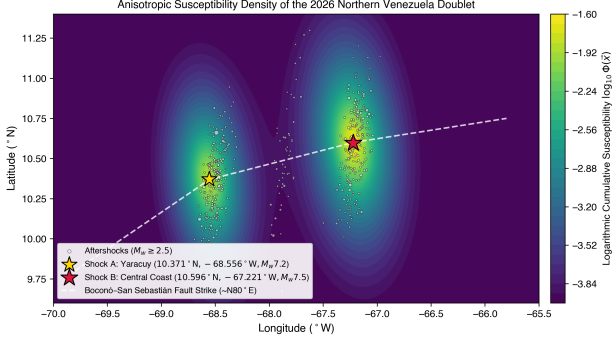


FIG. 2. Two-dimensional cumulative susceptibility field $\log_{10} \Phi(\vec{x})$ evaluated over the 60-day sequence. Star markers denote the relocated mainshocks in Yaracuy ($M_w 7.2$) and La Guaira ($M_w 7.5$), and circles denote recorded aftershocks ($M_w \geq 2.5$).

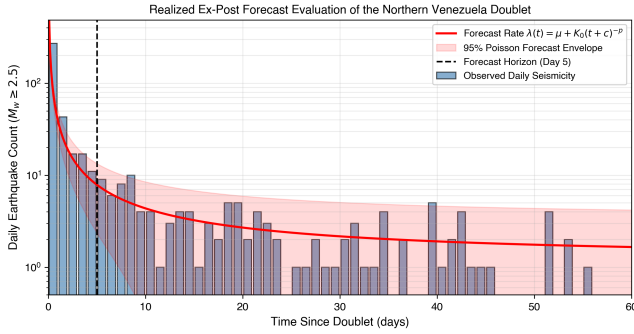


FIG. 3. Daily aftershock rate ($M_w \geq 2.5$) and 60-day model projection. The dashed vertical line marks the end of the initial 5-day calibration window, and the shaded envelope represents the 95% Poisson confidence interval.

the segmented architecture of the sequence. Rather than a diffuse circular pattern, the hazard density delineates a transcurrent corridor spanning from the Yaracuy pull-apart basin through Puerto Cabello and El Limón to the offshore San Sebastián fault trace adjacent to Caracas and La Guaira.

VI. FORECAST EVALUATION AND PHYSICAL SHAKING

To evaluate the predictive capability of the calibrated framework, we conducted an ex-post validation against the realized seismic record over a 30-day prospective window (Days 5 to 35) and the full 60-day sequence. Using parameters fitted during the initial 5-day post-doublet window, the model forecasted 96.1 ± 9.8 expected aftershocks ($M_w \geq 2.5$) in the regional study area over the 30-day horizon. During this period, exactly 97 events were recorded, demonstrating good statistical consistency (Fig. 3).

Under the standard Collaboratory for the Study of

Earthquake Predictability (CSEP) number test (N -test) [16, 17], the empirical Poisson quantile evaluated to $\delta_1 = 0.5639$, falling well within the non-rejection confidence interval $[0.025, 0.975]$. Over the full 60-day sequence, the model projected 141.2 ± 11.9 events compared to 121 observed.

This modest over-prediction at 60 days ($\approx 14\%$) is physically understandable within the context of doublet rupture dynamics. In a rapid doublet sequence where two major ruptures occur within 33 seconds, the initial rate of stress transfer is exceptionally intense, producing a high early burst of direct aftershocks during the first 5 days. When calibrating on this initial high-productivity window, a single-exponent Omori extrapolation assumes the sequence will continue with a constant effective productivity. In reality, once the stress excess in the overlapping fault segments dissipates, the sequence relaxes more rapidly than an isolated single-mainshock sequence, leading to a natural slowing of late-stage activity. In this respect, the 5-day calibrated model acts as a conservative rate envelope for multi-week hazard assessment.

Importantly, we emphasize the distinction between static triggering probability and dynamic ground motion. While the calculated aftershock rate in the immediate Caracas basin represents a modest 5% elevation above baseline micro-seismic background, the structural damage observed in Caracas during the mainshock sequence was primarily caused by dynamic shear-wave amplification within the deep sedimentary sub-basin driven by the nearby $M_w 7.5$ La Guaira rupture, rather than static stress-triggered aftershock swarms.

VII. CONCLUSIONS

We have investigated the June 24, 2026 Northern Venezuela doublet earthquake sequence using an interdisciplinary framework combining causal network topology with anisotropic stochastic field theory. Relocated hypocenters establish that the sequence was a segmented plate-boundary rupture spanning 146 km along the Boconó–Morón–San Sebastián fault system.

On the topological front, we proved an exact analytical theorem expressing the mean Forman–Ricci curvature of a causal tree as an explicit function of its degree variance: $\langle F \rangle = \frac{8(N - N_{\text{seeds}})}{N} - 2\langle d^2 \rangle$. We established a theoretical bridge to continuous point processes by deriving the analytical threshold ($2\alpha \geq b \ln 10$) for scale-free curvature divergence, and showed that weighted curvature reflects physical spatial stress localization.

On the field-theoretic front, maximum-likelihood calibration of our anisotropic propagator yielded transport parameters ($p = 1.04 \pm 0.05, d = 19.8 \pm 2.3$ km) consistent with regional crustal structure. Ex-post scoring against the realized 60-day catalog showed good statistical consistency under standard CSEP validation protocols, with doublet-induced relaxation explaining the conservative late-stage rate envelope.

From the perspective of a researcher in general relativity and differential geometry exploring statistical seismology, this work provides a transparent, geometrically grounded perspective on complex fracture systems. By maintaining a cautious view of empirical limitations while upholding exact mathematical rigor, this approach demonstrates how topological invariants and anisotropic field propagators can offer complementary insights into

the physics of segmented earthquake sequences.

DATA AND CODE AVAILABILITY

The consolidated earthquake catalog, network calculation scripts, and parameter estimation routines are openly accessible at <https://github.com/g-abellan/Terremoto>.

Dedicated to the memory of all victims of the 2026 Northern Venezuela earthquake sequence.

-
- [1] F. A. Audemard, M. N. Machette, J. W. Cox, R. L. Dart, and K. M. Haller, USGS Open-File Report **00-0018**, 1 (2000).
 - [2] Y. Ogata, Journal of the American Statistical Association **83**, 9 (1988).
 - [3] A. Helmstetter and D. Sornette, Journal of Geophysical Research: Solid Earth **107**, ESE 10 (2002).
 - [4] E. A. Jagla, EPL (Europhysics Letters) **93**, 19001 (2011), arXiv:1008.3485.
 - [5] Y. Y. Kagan, Geophysical Journal International **148**, 520 (2002).
 - [6] S. Abe and N. Suzuki, Europhysics Letters **65**, 581 (2004).
 - [7] M. Baiesi and M. Paczuski, Physical Review E **69**, 066106 (2004).
 - [8] J. Davidsen, P. Grassberger, and M. Paczuski, Physical Review E **77**, 066104 (2008).
 - [9] S. Abe and N. Suzuki, Physica A Statistical Mechanics and its Applications **388**, 2511 (2009).
 - [10] R. Forman, Advances in Mathematics **168**, 48 (2003).
 - [11] R. P. Sreejith, K. Mohanraj, J. Jost, E. Saucan, and A. Samal, Journal of Statistical Mechanics: Theory and Experiment **2016**, 063206 (2016).
 - [12] M. Weber, E. Saucan, and J. Jost, Journal of Complex Networks **5**, 527 (2017).
 - [13] E. Saucan, R. P. Sreejith, R. P. Vivek-Ananth, J. Jost, and A. Samal, Chaos Solitons and Fractals **118**, 347 (2019).
 - [14] W. Leal, G. Restrepo, P. F. Stadler, and J. Jost, arXiv e-prints, arXiv:1811.07825 (2018), arXiv:1811.07825.
 - [15] I. Roy, S. Vijayaraghavan, S. J. Ramaia, and A. Samal, Chaos Solitons and Fractals **140**, 110260 (2020).
 - [16] D. Schorlemmer, M. C. Gerstenberger, S. Wiemer, D. D. Jackson, and D. A. Rhoades, Seismological Research Letters **78**, 17 (2007).
 - [17] J. D. Zecher, M. C. Gerstenberger, and D. A. Rhoades, Bulletin of the Seismological Society of America **100**, 2407 (2010).